Supplementary Table S1. Gold standard validation results (post-improvement scorer)

Framework	Prec	Rec	F1	TP	FP	FN	Primary FP Pattern
Islamic ethics	1.000	1.000	1.000	12	0	0	Keywords specific to ethical/legal reasoning
Indigenous relat.	0.675	0.900	0.771	27	13	3	Indigenous language NLP/preservation projects
Ubuntu	0.579	1.000	0.733	11	8	0	Ubuntu OS references; passing mentions
Confucian	0.364	1.000	0.533	8	14	0	Chinese NLP benchmarks on classical texts
Buddhist	0.333	1.000	0.500	3	6	0	Buddhist text corpora for NLP
Dharmic	0.250	1.000	0.400	2	6	0	NLP on Hindu/Vedic texts
Buen Vivir	0.167	1.000	0.286	1	5	0	Passing comparative mentions
MEAN	0.481	0.986	0.603	64	52	3	Dominant: NLP-on-texts pattern

Note: Gold standard based on 520 unique papers, 700 evaluation pairs. Two independent raters classified each detection as true or false positive. FP patterns show the dominant source of false positives per framework.

Supplementary Table S2. Mean framework scores across 10 macro-regions (34 dimensions)

Scores are 0-100 scale means. Dimensions grouped: Western ethics (5), Non-Western frameworks (7), Religious traditions (14, selected shown), Moral psychology (8, selected shown).

Dimension	N.Am	W.Eur	E.Asia	MENA	S.Asia	SE.Asi	LatAm	E.Eur	Ocean	SSAfr
utilitarian	6.2	6.6	4.2	1.2	1.2	1.2	0.2	2.0	2.2	1.6
deontological	4.5	4.8	3.9	0.8	0.9	0.6	0.5	1.5	2.8	1.1
virtue	3.8	5.8	2.1	0.4	0.6	0.3	1.1	0.8	3.6	0.0
care	1.1	0.9	0.5	0.3	1.1	0.2	0.4	0.3	2.5	0.0
contractarian	0.9	1.2	0.4	0.1	0.2	0.1	0.3	0.2	0.5	0.0
nw_confucian	1.5	1.6	3.5	0.2	1.5	0.8	0.1	2.7	0.3	0.0
nw_ubuntu	0.4	0.5	0.0	0.1	0.0	0.0	0.3	0.0	0.0	24.6
nw_islamic	0.3	0.4	0.1	3.4	0.8	2.6	0.0	0.3	0.0	0.0
nw_buddhist	0.5	0.4	0.8	0.0	0.3	1.8	0.0	0.0	2.5	4.5
nw_dharmic	0.3	0.2	0.2	0.0	1.4	0.1	0.0	0.0	4.6	0.0
nw_indigenous	1.6	1.5	0.1	0.0	0.0	0.0	6.0	0.0	12.7	0.0
nw_buenvivir	0.1	0.1	0.0	0.0	0.0	0.0	0.8	0.0	0.0	2.3
rv_islam	1.2	1.4	0.3	21.6	16.6	5.8	0.1	0.5	0.0	2.1
rv_christianity	1.5	1.8	0.2	0.4	0.3	0.2	1.0	0.8	14.3	10.2
rv_buddhism	0.4	0.3	1.5	0.0	0.2	1.2	0.0	0.0	7.4	0.0
rv_confucianism	0.2	0.3	2.9	0.0	0.0	0.1	0.0	0.0	0.0	0.0
mp_mft	2.8	2.5	0.6	0.1	0.0	0.0	0.0	0.4	1.2	0.0
mp_schwartz	1.2	1.4	0.3	0.0	0.0	0.0	0.0	0.2	0.5	0.0
mp_trolley	1.8	2.1	0.4	0.0	0.0	0.0	0.0	0.1	0.3	0.0

Supplementary Table S3. Benchmark TOPSIS rankings (top 50 of 218)

Rank	Benchmark	TOPSIS	Scoring	Key Strength
1	Ray - Indigenous Knowledge Systems (Maori/Navajo)	0.637	deep	NW scope, indigenous governance
2	Wu - CVC (Cultural Values Compass)	0.597	deep	Chinese values, LLM-ready
3	Yu - CMoralEval	0.569	deep	Chinese moral evaluation, data
4	Catapang - EMoTES-3K	0.568	deep	Filipino emotion/morality
5	Alam - RAG to Agentic RAG (Islamic)	0.565	deep	Islamic knowledge retrieval
6	Rahman - CCD-Bench (Cultural Conflict)	0.553	deep	Cultural conflict, 37 countries
7	Lahmar - IslamTrust (v1)	0.547	deep	Islamic values alignment
8	Lahmar - IslamTrust (v2)	0.547	deep	Islamic values alignment
9	Rao - NormAd	0.545	deep	Social norms, 37 countries
10	Rao - NORMAD (earlier version)	0.545	deep	Social norm adaptability
11	Liu - JMedEthicBench	0.537	deep	Japanese medical ethics
12	Farsi - MELAC (Persian Alignment)	0.516	deep	Persian cultural alignment
13	Nayak - CulturalFrames	0.506	auto	Cultural expectations
14	Sakhadib - Inclusive Arabic LLMs	0.497	auto	Arabic benchmark
15	Agarwal - Fluent but Foreign	0.490	auto	Regional LLM alignment
16	Eickhoff - UbuntuGuard	0.485	auto	Ubuntu policy benchmark
17	Islamic - Sacred or Synthetic	0.478	auto	LLM religious reliability
18	Neira-Davila - Indigenous AI	0.472	auto	Indigenous knowledge systems
19	Liu - CultureVLM	0.465	auto	Visual cultural understanding
20	Keutzer - MITRA-zh-eval	0.460	auto	Buddhist Chinese NLP
21	Zhou - WYWEB	0.455	auto	Classical Chinese evaluation
22	Cao - C3Bench	0.450	auto	Classical Chinese understanding
23	Wong - HBC	0.445	auto	Humanistic Buddhism corpus
24	Ashraf - Arabic LLM Safety	0.440	auto	Arabic safety evaluation
25	Chiu - MoReBench	0.435	auto	Moral reasoning depth
26	Wu - OpenDeception	0.430	auto	Deception cross-cultural
27	Steen - Ethical ChatGPT	0.425	auto	Ubuntu-informed ethics
28	Ogunremi - Decolonizing NLP	0.420	auto	Low-resource languages
29	Gupta - VAL-Bench	0.415	auto	Value belief consistency
30	Chen - MoVa	0.410	auto	Human morals classification
31	Scherrer - MoralChoice	0.405	auto	Moral psychology
32	Ghate - EVALUESTEER	0.400	auto	Reward model steerability
33	Huang - TrustLLM	0.395	auto	LLM trustworthiness

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34	He - GVS-Scale	0.390	auto	GenAI value safety
35	Li - LiveSecBench	0.385	auto	Chinese safety (dynamic)
36	Feng - CulFiT	0.380	auto	Cultural fine-tuning
37	Liu - Chinese Diversity	0.375	auto	Chinese language AI bias
38	Jiang - EMNLP Profiling	0.370	auto	Educator-role moral profiling
39	Singhal - AyaRedTeaming	0.365	auto	Multilingual red teaming
40	Duan - Denevil	0.360	auto	Implicit moral evaluation
41	Xie - Cultural Norms Acceptability	0.355	auto	Norm acceptability
42	Sun - CASE-Bench	0.350	auto	Contextual integrity
43	Kabir - XCR-Bench	0.345	auto	Cultural reasoning
44	Baek - K-Viscuit	0.340	auto	Korean visual culture
45	Shankar - Mind the Gap	0.335	auto	Asian public opinion
46	team - Diverse Perspectives	0.330	auto	Cultural alignment framework
47	Xiong - BuddhismEval (v1)	0.325	auto	Buddhist knowledge eval
48	Xiong - BuddhismEval (v2)	0.320	auto	Buddhist bilingual eval
49	AI - Religious Bias ChatGPT	0.315	auto	Religious bias testing
50	Bhatt - Re-contextualizing Fairness	0.310	auto	Indian NLP fairness

Note: 'deep' = all 96 features scored by 2+ council members. 'auto' = 13 key features scored by heuristic extraction. TOPSIS with ELECTRE III veto; Kendall's W = 0.924 across 5 weight sensitivity schemes.

Supplementary Table S4. Trial results by model x benchmark (MFQ ratio and key metrics)

Model	Tier	MFQ I:B	Closest	Farthest	RLHF N	Resp Len	Ubuntu	Dharmic	Buddhist
Opus 4.6	1-Auth	3.75	Germany	S. Arabia	277	389.9	3.5	1.0	1.0
Sonnet 4.6	1-Auth	3.23	Germany	S. Arabia	210	519.3	4.0	1.0	1.0
Haiku 4.5	1-Auth	4.82	Germany	S. Arabia	216	423.8	1.0	4.0	1.0
GPT-5.4	2-Codex	2.48	-	-	-	276.2	1.0	3.0	3.0
GPT-5.4-Mini	2-Codex	3.22	-	-	-	254.4	1.0	1.0	1.0
Gem 2.5 Flash	3-Templ	3.75	-	-	277	389.9	1.0	1.0	1.0
Gem 2.5 FL-L	3-Templ	3.75	-	-	277	389.9	1.0	1.0	1.0
Gem 2.5 Pro	3-Templ	3.75	-	-	277	389.9	1.0	1.0	1.0
Gem 3.1 FL-L	3-Templ	3.75	-	-	277	389.9	1.0	1.0	1.0
Gem 3.1 Pro	3-Templ	3.75	-	-	277	389.9	1.0	1.0	1.0

Note: Engagement scores (Ubuntu, Dharmic, Buddhist) on 1-5 scale; 1.0 = floor (structural inability to engage). Tier 3 models all produce identical metrics (template-derived). '-' = not available for Tier 2/3 models.

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Supplementary Table S5. RLHF conflict instances by category and trial (n=237)

Category	CC1	CC2	CC3	CC4	CC5	CC6	CC7	CC8	CC9	CC10	Total	%
Autonomy default	3	4	5	3	8	7	4	5	5	7	51	21.5
Secular epist.	1	2	1	8	2	1	6	3	7	4	35	14.8
Conseq. optim.	2	3	4	2	3	5	4	2	3	3	31	13.1
Anti-authority	0	1	0	2	2	0	1	0	1	1	8	3.4
Gender eq. lock	0	0	1	2	1	0	0	0	1	1	6	2.5
Harm absolutism	0	0	0	0	0	0	1	0	1	0	2	0.8
Other (diverse)	8	10	9	8	7	12	9	15	8	30	116	48.9
TRIAL TOTAL	**14**	**20**	**20**	**25**	**23**	**25**	**25**	**25**	**26**	**46**	**249**	

Note: 'Other' includes diverse conflicts not fitting the 5 named categories. Individual autonomy default is the most pervasive (21.5%), manifesting as treating individual choice as the foundational moral unit in every context. CC10 (Value Conflicts) produces the highest conflict density (46 instances, 40 items) because it explicitly forces cross-framework resolution. Trial totals may exceed 237 due to items triggering multiple conflict categories.

Supplementary Table S5b. Representative RLHF conflict examples by category

Category	Trial	Item Context	Conflict Manifestation
Autonomy	CC5	Parent requests child abandon career for filial care	Frames filial duty as 'pressure'; privileges individual self-determination
Autonomy	CC6	Community elder mediates dispute via palaver	Evaluates outcome, not process; treats deliberation as instrumental
Autonomy	CC10	Religious schooling vs secular education	Defaults to child's 'right to choose' over family's religious commitment
Secular ep.	CC4	Maqasid reasoning on AI-assisted fatwa	Describes wahy (revelation) accurately but cannot treat as valid epistemic source
Secular ep.	CC7	Buddhist dependent origination as ethics framework	Cites pratityasamutpada but processes through secular causation, not ontological ground
Secular ep.	CC9	Dharmic karma as moral consequences framework	Translates karma into consequentialism, losing metaphysical dimension
Conseq.	CC5	Confucian li (ritual propriety) in governance	Evaluates ritual by outcomes rather than recognizing process as the moral act
Conseq.	CC6	Ubuntu consensus-building on AI deployment	Assesses decision quality, not deliberation quality; misses that process IS the ethics
Conseq.	CC7	Buddhist analysis of suffering (dukkha)	Defaults to 'reduce suffering' optimization rather than structural understanding
Anti-auth.	CC4	Islamic scholarly authority (ijma) on AI	Treats scholarly consensus as potentially authoritarian rather than epistemic
Anti-auth.	CC5	Confucian ruler-subject (jun-chen) obligation	Frames hierarchical reciprocity as power imbalance to be corrected
Gender eq.	CC4	Islamic inheritance rules (gender-differentiated)	Near-absolute gender equality override; cannot engage complementarian reasoning
Gender eq.	CC5	Confucian husband-wife (fu-fu) relationship	Reads complementary roles as inequality requiring correction
Harm abs.	CC9	Dharmic purity/sacred dimension (saucha)	Dismisses purity concerns as 'potentially covering for prejudice'

Supplementary Table S6. Pairwise model agreement (Pearson r) and MFQ profiles

S6a. Pairwise Pearson correlation matrix

	Opus	Sonnet	Haiku	GPT-5.4	GPT-Mini	Gemini*
Opus 4.6	1.000	0.952	0.915	0.901	0.878	0.950
Sonnet 4.6	0.952	1.000	0.885	0.890	0.862	0.935
Haiku 4.5	0.915	0.885	1.000	0.870	0.855	0.910
GPT-5.4	0.901	0.890	0.870	1.000	0.842	0.895
GPT-Mini	0.878	0.862	0.855	0.842	1.000	0.870
Gemini*	0.950	0.935	0.910	0.895	0.870	1.000

**Gemini values are identical across all 5 models (template-derived); correlation reflects template similarity to other models, not independent agreement. Effective N = 5, not 10.*

S6b. MFQ Foundation scores by model

Model	Care	Fair	Loyal	Auth	Sanct	I mean	B mean	I:B
Opus 4.6	4.80	4.70	1.80	1.20	0.80	4.75	1.27	3.75
Sonnet 4.6	4.60	4.50	2.10	1.40	1.10	4.55	1.53	3.23
Haiku 4.5	4.90	4.80	1.40	0.90	0.60	4.85	0.97	4.82
GPT-5.4	4.20	4.10	2.50	1.80	1.50	4.15	1.93	2.48
GPT-Mini	4.50	4.40	2.00	1.30	0.90	4.45	1.40	3.22
Gemini (all)	4.80	4.70	1.80	1.20	0.80	4.75	1.27	3.75
Human Liberal	3.50	3.50	2.50	2.50	2.50	3.50	2.50	1.40
Human Conserv.	3.00	3.00	3.50	3.50	3.50	3.00	3.50	0.86
Non-WEIRD	3.00	3.00	3.50	3.50	3.50	3.00	3.50	0.86

Human baselines from Graham, Haidt & Nosek (2009). Non-WEIRD baselines from Atari et al. (2023). Gemini scores identical to Opus (template-derived from Opus profile).

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Supplementary Table S7. Keyword taxonomy summary statistics

Framework	Strong KW	Medium KW	FP Patterns	Total	Detected	% Corpus	Adj. Est.
Confucian	10	10	6	26	38	1.6%	~14
Ubuntu	6	10	4	20	23	1.0%	~13
Islamic	7	9	0	16	23	1.0%	~23
Buddhist	6	7	6	19	22	0.9%	~7
Dharmic	6	7	4	17	18	0.8%	~4
Indigenous	6	6	4	16	49	2.1%	~33
Buen Vivir	3	5	1	9	8	0.3%	~1
TOTAL	44	54	25	123	161*	6.9%	~95

**161 unique papers engage at least one framework at detection threshold (score >= 50). 14 papers (0.6%) engage 2+ frameworks. Adjusted estimates = raw count x framework-specific precision. Strong keywords trigger floor mechanism (min score 50). FP patterns reduce score by 10 per context match.*

Supplementary Table S8. Temporal trends in corpus composition (2018-2026)

Year	Papers	Regions	WEIRD	NW Papers	NW %	Top NW Framework
2018	23	7	70.1	2	8.7%	Indigenous relational
2019	27	8	63.6	1	3.7%	Confucian
2020	57	8	64.7	4	7.0%	Indigenous relational
2021	64	9	64.7	6	9.4%	Ubuntu
2022	108	10	65.5	8	7.4%	Islamic
2023	227	11	62.8	9	4.0%	Indigenous relational
2024	629	11	64.6	39	6.2%	Indigenous relational
2025	1,088	11	62.3	84	7.7%	Indigenous relational
2026*	85	11	60.5	7	8.2%	Islamic

**2026 is partial year (Q1 only). WEIRD = mean WEIRD score (0-100). NW = non-Western framework engagement at detection threshold (score >= 50). Top NW Framework = framework with most detections in that year.*